

A qualitative strengthening of the three-halves endpoint bound

Research proof: independently reviewed

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Status. Complete proof, independently reviewed by the root and rational-design research agents. Its endpoint-profile and variance-sensitive Laguerre inputs have also been independently reviewed. This qualitative result is superseded by the stronger near-quadratic bootstrap; it is retained as a separate structural argument. No explicit exponent exceeding $3/2$ is claimed in this note.

Let a sequence of monotone uniform-interval comparison tensors have $m \rightarrow \infty$ columns, arbitrary positive row weights, and $\mathbb{E}_w \prod_{j \in S} p_j = 1/(|S| + 1)$. Put $u_j = 1 - p_j$, $s = \sum_j u_j$, $\ell = \lfloor m/4 \rfloor$, $N = m - \ell$, $\alpha = \ell/m$, and $z_j = \ell u_j$. Define the row range

$$H_m = \max_j z_j - \min_j z_j.$$

The previously reviewed uniform endpoint-profile theorem, combined with finite mean-deficit balance, gives

$$\sup_{s \leq L} H_m \longrightarrow 0 \quad (L < \infty \text{ fixed}). \quad (1)$$

Indeed, for $s \leq L$ the finite bound gives $\max u_j = O_L(1/m)$, so all entries are positive and $\lambda = \sum u_j/(1 - u_j) = s + O_L(1/m)$. The endpoint profile then states uniformly that $mu_j = s + o(1)$. The same finite bound gives globally

$$H_m \leq C(s + 1), \quad \mathbb{E}_w (s + 1)^a e^{-cs} \leq C_{a,c}/m \quad (2)$$

for every fixed $a \geq 0, c > 0$.

We use the mixture $\mu(q) = (\ell + 1)w_q \mathbb{E}_B A_B(q)$, $A_B = \prod_{j \in B} p_j$, where B is a uniform ℓ -subset, and $x = \alpha s$. Exact polarization expectations equal the beta law $b_\ell(z) = (\ell + 1)\ell^{-1}(1 - z/\ell)^\ell$ on $[0, \ell]$.

Lemma 1 (Range-sensitive comparison). *Fix $\kappa > 0$ and put $d = \lfloor \kappa\sqrt{m} \rfloor$. Let $Q_a(z) = \sum_{j=0}^d c_{aj} L_j(\theta z)$ for a fixed sufficiently small $\theta > 0$, and let $B_a = \sum_j |c_{aj}|$, $B = B_1 B_2$. Uniformly over these coefficients,*

$$|\mathbb{E}_\mu Q_1(x) Q_2(x) - \beta_\ell(Q_1 Q_2)| = o(Bd/m). \quad (1)$$

Proof. We detail why each formerly $O(Bd/m)$ error acquires a little-oh factor. For a complementary subset R , let $v = N^{-1} \sum_{j \in R} z_j$. The centered variance satisfies the sharper estimate

$$\sigma^2 = \sum_{j \in R} (z_j - v)^2 \leq N H_m v. \quad (2)$$

To see this, write $a = \min_{j \in R} z_j$ and $b = \max_{j \in R} z_j$. The variance is at most $(b - v)(v - a) \leq H_m v$ times N .

The all-complex Cauchy estimate and the off-diagonal Gaussian Laguerre bound give respectively

$$\frac{|e_k(z - v)|}{\binom{N}{k}} \leq (\sqrt{e} \sigma \sqrt{k}/N)^k, \quad \frac{|(Q_1 Q_2)^{(k)}(v)|}{k!} \leq B e^{\theta v} (2\sqrt{\theta d/v})^k / k!.$$

At $v = 0$ the centered error vanishes identically. For $k = 2$, using (2), the normalized integrated error is at most

$$CB \frac{d}{m} m \mathbb{E}_w [H_m e^{-cs}] = o(Bd/m).$$

Here (1) handles $s \leq L$ and (2) makes the $s > L$ contribution uniformly small as $L \rightarrow \infty$. For all $k \geq 3$ we use the coarser $H_m \leq C(s + 1)$ and the factorial weighted-moment bound. As in the variance-sensitive polarization lemma, their total is

$$O\left(B(d/m)^{3/2}\right) = o(Bd/m).$$

This proves the improved polarization comparison.

For the mixture comparison, work first on $s \leq m/256$, where $p_i \geq 1/2$. Under the success-weighted subset law put $\Delta = v - x$. The exact bias formula and the identity $\sum_j (u_j - s/m)^2 \leq (\max u_j - \min u_j)s$ give

$$0 \leq \mathbb{E}\Delta \leq C H_m x / m. \quad (3)$$

For the variance, the subset size is fixed. Thus subtracting $a = \min_j z_j$ from every sampled coefficient leaves its variance unchanged. The remaining

coefficients are nonnegative, so the negative pairwise inclusion covariances imply

$$\text{Var } \Delta \leq N^{-2} \sum_j (z_j - a)^2 \leq N^{-2} H_m \sum_j z_j \leq CH_m x / m. \quad (4)$$

Consequently $\mathbb{E} \Delta^2 \leq CH_m x / m + CH_m^2 x^2 / m^2$.

The off-diagonal derivative bounds are $|P'(x)| \leq CB \sqrt{d/x} e^{\theta x}$ and $|P''(z)| \leq CB d e^{\theta z} / z$ for $P = Q_1 Q_2$. The integral Taylor remainder obeys

$$|P(v) - P(x) - P'(x)(v - x)| \leq CB d e^{\theta \max(x, v)} \frac{(v - x)^2}{x} \quad (5)$$

for $x > 0$, including $v = 0$. Indeed, with $v = x + \Delta$, $(1 - t)/(x + t\Delta) \leq 1/x$ for $0 \leq t < 1$ and $v \geq 0$. At $x = 0$ all deficits and all errors vanish. Combining (3)–(5), averaging with $\bar{A} \leq e^{-\alpha s}$, and normalizing yields a bound of the form

$$CB \frac{d}{m} m \mathbb{E}_w [(H_m(1 + s) + H_m^2(1 + s)/m) e^{-cs}] = o(Bd/m)$$

by (1)–(2). The region $s > m/256$ has exponentially small error by the previously established crude uniform-subset comparison, even for $d = O_\kappa(\sqrt{m})$. This proves (1). $\square$

Theorem 1. *Under the tensor hypotheses above,*

$$m^{3/2} \max\{w_1, w_K\} \longrightarrow 0.$$

In particular the exceptional-die optimum satisfies $g(n)/n^{3/2} \rightarrow \infty$.

Proof. Use the same scaled Laguerre taper $T_d(y) = \sum_{j < d} (1 - j/d) L_j(y)$ and $R_d(z) = (2/(\theta d) - z) T_d(\theta z)^2$. Its beta expectation is bounded below by a fixed positive constant. The recurrence identity $y T_d = d^{-1} \sum_{j < d} L_j - L_d$ factors R_d into Laguerre sums with coefficient norms $O(d)$ and $O(1)$. Lemma 1 therefore gives an error $o(d^2/m) = o_\kappa(1)$ for every fixed κ . Hence the last row satisfies $x_K < 2/(\theta d)$ for all sufficiently large m .

The normalized Christoffel square at this row has coefficient norm $O(1)$, value one there, and beta integral $O(1/d)$. Lemma 1 gives

$$\mu(K) \leq C/d + o(d/m) = \frac{C}{\kappa \sqrt{m}} + o_\kappa(m^{-1/2}).$$

Also $s_K = O(1/d)$, so every $A_B(K) \geq 1 - s_K \geq 1/2$ eventually. Thus

$$\limsup_{m \rightarrow \infty} m^{3/2} w_K \leq C/\kappa.$$

Since $\kappa > 0$ was arbitrary, the limit is zero. Reversal and complementation give the first-row statement. Equal weights give the asserted consequence for dice. $\square$